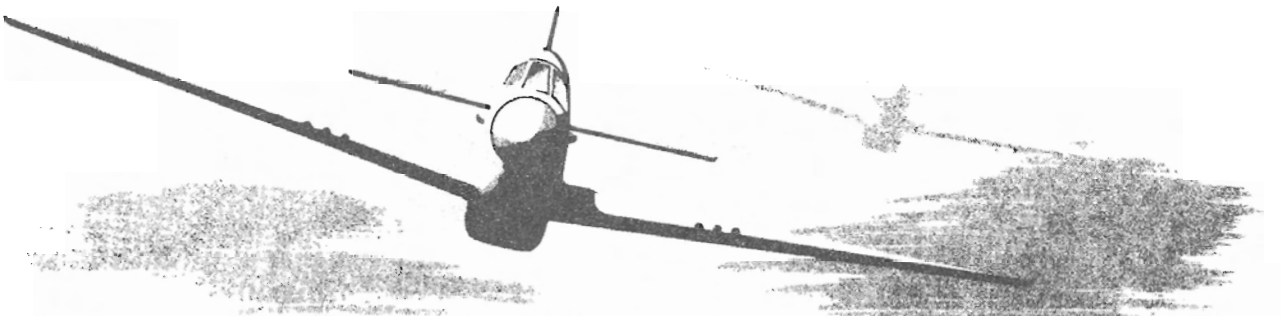


AIRSPEED INSTALLATION CORRECTION

APPLY CORRECTION TO INSTRUMENT READING TO OBTAIN CALIBRATED AIRSPEED

GEAR AND FLAPS UP—CANOPY CLOSED

MPH	CORRECTION	MPH	CORRECTION
100	5	60	6
120	5	90	3
140	4	100	1
160	4	110	−1
180	3	120	−2
200	3	130	−3
220	2	150	−4
240	2	170	−4
260	2		
280	2		
300	2		
350	3		
400	4		

AIRSPEED COMPRESSIBILITY CORRECTION

SUBTRACT CORRECTION FROM CALIBRATED AIRSPEED TO OBTAIN EQUIVALENT AIRSPEED

PRESSURE ALTITUDE	CALIBRATED AIRSPEED—MPH							
	150	200	250	300	350	400	450	500
5,000	0	0	1	1	2	3	4	5
10,000	0	1	2	3	4	6	8	
15,000	0	1	3	4	7	9		
20,000	1	2	4	6	10	14		
25,000	1	3	5	9	13			
30,000	2	4	7	12	18			
35,000	2	5	10	16				

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Figure A-1

FREE AIR TEMPERATURE *Correction Chart*

ALTITUDE—FEET	INDICATED AIRSPEED—MPH				
	150	200	250	300	320
0		4	6	8	9
5,000		4	7	9	11
10,000		5	8	11	12
15,000	4	6	9	13	15
20,000	5	8	11	16	18
25,000	6	9	13	19	21
30,000	7	11	17	22	24
35,000	8	13	20		
40,000	10	17	24		
45,000	12	21			

SUBTRACT CORRECTION SHOWN FROM CARBURETOR AIR TEMPERATURE TO OBTAIN FREE AIR TEMPERATURE IN DEGREES CENTIGRADE.

DATA BASIS: FLIGHT TEST
DATA AS OF: 9-2-53

109-93-1745

Figure A-2

FREE AIR TEMPERATURE CORRECTION.

Since no free air temperature gage is provided in this airplane, a chart for converting indicated carburetor air temperature to free air temperature is given in figure A-2. The corrected free air temperature can be used with calibrated airspeed to obtain true airspeed.

EXAMPLE—USE OF CORRECTION TABLES.

An airplane is flying at 25,000 feet pressure altitude. The indicated carburetor air temperature is −15°C, and the indicated airspeed reading is 300 mph. What is the true airspeed?

Airspeed indicator reading (IAS)	300 mph
Correction for installation error	2 mph
Calibrated airspeed (CAS)	302 mph

Indicated carburetor air temperature	−15°C
Correction to obtain free air temperature	−19°C
Corrected free air temperature	−34.0°C

Use these values of CAS and free air temperature with a Type D-4 or Type G-1 airspeed computer to determine the true airspeed of 438 mph.

When a Type AN5835-1 dead-reckoning computer is used, CAS usually must be corrected for compressibility error.

Calibrated airspeed (CAS)	302 mph
Compressibility Error	−9 mph
Equivalent Airspeed (EAS)	293 mph

Use this value of EAS with the dead-reckoning computer to determine the true airspeed of 438 mph.

TAKE-OFF DISTANCES.

The Take-off Distances charts (figure A-4) give take-off ground-run distances and total distances to clear a 50-foot obstacle. The charts are tabulated for several different gross weights, altitudes, and temperatures on a hard-surface runway. Distances given are for standard flaps-up take-offs. For a minimum-run take-off, refer to Section II.

CLIMB.

Best climb speed, fuel consumption, time to climb, and rate of climb (using Military Power or Normal Power) can be determined for different configurations from the Military Power and Normal Power Climb charts (figures A-5 and A-6). A fuel allowance for warm-up, taxi, and take-off is listed in the column labeled "SEA